Exercise: Data Cleaning

Learn how to clean and prepare raw data by removing missing credit scores, filling missing incomes for specific groups, and eliminating duplicate records. This lesson helps you apply data cleaning techniques essential for transforming datasets within ETL processes to ensure reliable analysis.

In this exercise, we're given a dataset about the company's customers. Unfortunately, the data is not clean. There are a few missing and duplicate values. Our task is to clean the data and prepare it for further analysis. Let's take a look at the data.

main.cpp C++

```
+---------+----+---+------+------+------------+
|id_number|name|age|gender|income|credit_score|
+---------+----+---+------+------+------------+
|   135245|John| 25|     M| 50000|        null|
|   223452|Jane| 30|     F| 60000|         700|
|   341412| Joe| 35|     M| 70000|         800|
|   341412| Joe| 35|     M| 70000|         800|
|   574355|Jill| 40|     F| 80000|         900|
|   253774|Jack| 45|     M| 90000|        1000|
|   856585|Jack| 75|     M|  null|         300|
+---------+----+---+------+------+------------+
```

Customer data

We can see that there are a few problems with the data:

- There are missing values in the `credit_score` and `income` columns.
- There are two records that are complete duplicates in the dataset.

> **Note:** Answer the quiz questions after each task to complete the exercise.

Remove the missing `credit_score` values

According to employees within the business, it is a well-known issue that the `credit_score` column can sometimes contain missing values for no apparent reason. Unfortunately, there is no way to recover the missing values without creating a bias in the data.

Our task is to remove these rows with missing `credit_score` from the dataset.

main.py Python 3.10.4

```
from pyspark.sql import SparkSession

# Create a Spark session
spark = SparkSession.builder.appName("MAR").getOrCreate()

# Load the CSV file
print('\nLoading the data')
data = spark.read.option('header', 'true').option('inferSchema', 'true')\
    .csv('/usercode/tmp/spark_customers.csv')

# Drop the rows with missing credit_score
data = data.dropna(subset=["credit_score"])
```

Run

Removing the missing credit_score values

Ask

Quiz

Technical Quiz

1. Which type of missing data is presented in the above code?

A. MAR

B. MCAR

C. MNAR

1 / 2

Submit Answer

Remove the missing `income` values

Next, we notice that there are rows with missing income. According to the business, older retired customers will sometimes have missing values as their income.

Since this data is a MAR value, instead of dropping these records, the business has decided to fill in the missing values for all customers older than 70 with a known income value for retired customers, which is 10000.

main.py Python 3.10.4

```
# Replacing missing income values if a customer is over 70
data = data.withColumn("income", when( (col("age") >= 70) & data["Income"

# Print data
data.show()
```

Ask

Run

Filling missing income values

Quiz

Technical Quiz

1. **The company has realized that filling in missing income values with 10000 was not a good idea.**

   They want to replace the missing values with something more realistic. Which approach would introduce the least bias into the data?

A. Replace missing values with the median of the income column

B. Replace missing values with the mean of the income column

C. Replace missing values with 20000 instead of 10000

D. Replace missing values with the median of the income column of all customers over 70 that are retired

1 / 1

Submit Ar

Ask

Duplicate records

It looks like two records are complete duplicates. We'll have to remove these from the dataset.

main.py Python 3.10.4

```
# Drop complete duplicates
data = data.drop_duplicates()

# Print data
data.show()
```

Run

Removing complete duplicates

Quiz

Technical Quiz

1. **What if the duplicate records were not complete duplicates and had the same values for all columns except `id_number`?**

   How would we handle the duplicates in this case?

Ask

A. Drop both of them.

B. Keep the first record.

C. Keep the last record.

D. Keep both records.

1 / 1

Submit Answer

Ask